MAP Cleaning Report

Assigned Column: MAP

Overview

For our breakout group, we were assigned the **MAP** column, which stands for **Mean Arterial Pressure**. MAP is a clinical value that represents the average pressure in a patient's arteries during one full heartbeat cycle. It is important because it gives an idea of whether enough blood pressure is being maintained to supply organs with oxygenated blood.

Although MAP was our assigned column, we could not clean it properly by only looking at the original MAP values. This is because MAP is a calculated value that depends on both **SBP** and **DBP**.

The formula used was:

MAP = (SBP + 2(DBP)) / 3

Because of this, we first cleaned **SBP** and **DBP**, then recalculated MAP using the filtered blood pressure values. Any remaining missing values in MAP were then filled using the median.

Section 8 Deliverables

The main deliverables for this cleaning task were to identify the assigned column, inspect the data, decide on a valid cleaning method, handle missing or invalid values, justify our choices, and explain the final cleaned result. Our final deliverable is a cleaned MAP column that is calculated from cleaned SBP and DBP values and has missing values handled using median imputation.

1. Inspection of the Data

We started by inspecting the **SBP**, **DBP**, and **MAP** columns. This helped us check the data type, missing values, unusual values, and the overall range of the data.

```
unique()
describe()
isnull().sum()
dtype
```

For SBP, we noticed that it was stored as an object type instead of a numeric type. This meant that before doing calculations, we had to convert it into a numeric column. For DBP, we also checked the data type, unique values, and missing values before applying the cleaning steps.

2. Clinical Meaning and Valid Ranges

SBP

SBP stands for systolic blood pressure. It is the top number in a blood pressure reading, such as 120/80. It represents the pressure in the arteries when the heart beats.

The valid SBP range we used was:

```
50-250 mmHg
```

Values below 50 or above 250 were treated as invalid because they are outside the expected clinical range for this dataset.

DBP

DBP stands for diastolic blood pressure. It is the bottom number in a blood pressure reading and represents the pressure in the arteries when the heart is resting between beats.

The valid DBP range we used was:

```
30-150 mmHg
```

Values outside this range were treated as invalid.

MAP

MAP stands for mean arterial pressure. It is calculated using SBP and DBP. The valid MAP range given in the assignment was:

```
40-180 mmHg
```

However, since MAP is calculated, we focused mainly on making sure that it was recalculated from cleaned SBP and DBP values rather than trusting the original MAP column.

3. Type Conversion

We converted the blood pressure columns to numeric values using **pd.to_numeric**. This was important because any non-numeric entries would be converted to missing values instead of causing errors in the code.

```
df["SBP"] = pd.to_numeric(df["SBP"], errors="coerce")
df["DBP"] = pd.to_numeric(df["DBP"], errors="coerce")
```

Using **errors="coerce"** allowed the code to safely handle messy values by turning them into **NaN**. This made the columns suitable for range filtering, imputation, and MAP calculation.

4. Flagging and Replacing Out-of-Range Values

After converting the columns to numeric form, we filtered values based on the valid clinical ranges.

For SBP:

```
df.loc[(df["SBP"] < 50) | (df["SBP"] > 250), "SBP"] = np.nan
```

For DBP:

```
df.loc[(df["DBP"] < 30) | (df["DBP"] > 150), "DBP"] = np.nan
```

We replaced invalid values with **NaN** instead of deleting the entire row because each row represents a patient. Removing rows could cause us to lose useful patient information from the rest of the dataset.

5. Imputation Method and Justification

We used **median imputation** to fill missing values after filtering.

For SBP:

```
sbp_median = df["SBP"].median()
df["SBP"] = df["SBP"].fillna(sbp_median)
```

For DBP:

```
dbp_median = df["DBP"].median()
df["DBP"] = df["DBP"].fillna(dbp_median)
```

We chose the median instead of the mean because blood pressure data can contain extreme values. The median is less affected by outliers, so it is a safer choice for clinical data.

After cleaning SBP and DBP, we recalculated MAP using the formula:

```
df["MAP"] = ((df["SBP"] + 2 * df["DBP"]) / 3).round(2)
```

As a final fallback, any remaining missing values in MAP were filled using the MAP median:

```
map_median = df["MAP"].median()
df["MAP"] = df["MAP"].fillna(map_median)
```

This ensured that the final MAP column had no missing values.

One small correction we noticed is that if the notebook print statement says “imputed with mean,” it should be changed to “imputed with median,” because the actual code uses the median value.

```
print(f"Imputed with median: {map_median:.2f} mmHg")
```

6. Why We Recalculated MAP Instead of Only Filtering It

We chose to recalculate MAP because MAP depends directly on SBP and DBP. If the original SBP or DBP values were dirty, then the original MAP could also be incorrect.

In our notebook, we found that some MAP values did not match the correct formula because they had been calculated from unclean blood pressure values. Therefore, simply filtering the existing MAP column would not fully solve the problem.

Recalculating MAP from the cleaned SBP and DBP values made the MAP column more consistent, accurate, and clinically meaningful.

7. Final Sanity Check

At the end, we ran a final validation check for SBP, DBP, and MAP. This check counted values outside the valid ranges and counted any remaining missing values.

```
checks = [('SBP', 50, 250), ('DBP', 30, 150), ('MAP', 40, 180)]

for col, lo, hi in checks:
n_invalid = ((df[col] < lo) | (df[col] > hi)).sum()
n_null = df[col].isnull().sum()
flag = 'PASS' if (n_invalid == 0 and n_null == 0) else 'WARNING'
print(f"{flag} {col}: range {df[col].min():.1f}-{df[col].max():.1f} | "
f"out-of-range: {n_invalid} | NaNs: {n_null}")
```

The final output was:

```
PASS SBP: range 55.0-250.0 | out-of-range: 0 | NaNs: 0
PASS DBP: range 30.0-150.0 | out-of-range: 0 | NaNs: 0
WARNING MAP: range 39.3-173.3 | out-of-range: 1 | NaNs: 0
```

This showed that SBP and DBP fully passed the range and missing-value checks. MAP had no missing values, but it had one value slightly below 40 mmHg. We kept this value because it was recalculated from valid SBP and DBP values, so it could represent a critically ill patient rather than a data entry error.

Conclusion

For this cleaning task, we cleaned the MAP column by first preparing the SBP and DBP columns, since MAP is calculated from both of them. We converted the blood pressure columns to numeric values, flagged out-of-range values, replaced invalid entries with NaN, and used median imputation to fill missing values.

After that, we recalculated MAP using the cleaned SBP and DBP values. This was a better approach than only filtering the original MAP column because it ensured that MAP was consistent with the correct clinical formula. Overall, our cleaned MAP column is more reliable, has no missing values, and is based on valid blood pressure data.